Energy and potential

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■ **Example**

Given the non-uniform field $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$, determine the work expended in carrying 2 C from $B(1,0,1)$ to $A(0.8,0.6,1)$ along the shorter arc of the circle $x^2 + y^2 = 1$, $z = 1$.

$d\mathbf{L} = dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z$ • Differential path, rectangular coordinate

$$W = -Q\int_B^A \mathbf{E}\cdot d\mathbf{L}$$

$$= -Q\int_B^A (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z)\cdot(dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z)$$

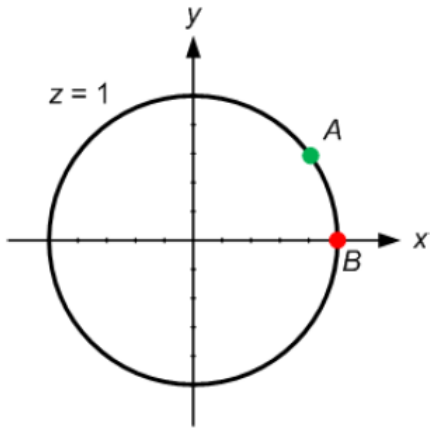
$$= -2\int_1^{0.8} y\,dx - 2\int_0^{0.6} x\,dy - 2\int_1^1 2\,dz$$

• Circle equation: $x^2 + y^2 = 1$

$$x = \sqrt{1-y^2}$$

$$y = \sqrt{1-x^2}$$

$$W = -2\int_1^{0.8} \sqrt{1-x^2}\,dx - 2\int_0^{0.6} \sqrt{1-y^2}\,dy - 2\int_1^1 2\,dz$$



$$= -2\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_1^{0.8} - 2\left[\frac{y}{2}\sqrt{1-y^2} + \frac{1}{2}\sin^{-1}y\right]_0^{0.6}$$

$$= \underline{\underline{-0.962\ \text{J}}}$$

$$\int \sqrt{a^2-u^2}\,du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\sin^{-1}\frac{u}{a}$$

■ Example

Redo the example, but use the straight-line path from *B* to *A*.

$$W = -Q\int_B^A \mathbf{E}\cdot d\mathbf{L}$$

$$= -Q\int_B^A (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z)\cdot(dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z)$$

Any two of the following three equations for planes passing through the line are sufficient to define the line:

$$y - y_B = \frac{y_A - y_B}{x_A - x_B}(x - x_B)$$

$$z - z_B = \frac{z_A - z_B}{y_A - y_B}(y - y_B)$$

$$x - x_B = \frac{x_A - x_B}{z_A - z_B}(z - z_B)$$

From the first equation we have

$$y = -3(x - 1)$$

and from the second we obtain

$$z = 1$$

$$W = -2\int_1^{0.8} y\,dx - 2\int_0^{0.6} x\,dy - 4\int_1^1 dz$$

$$= 6\int_1^{0.8} (x - 1)\,dx - 2\int_0^{0.6} \left(1 - \frac{y}{3}\right) dy$$

$$= -0.96\ \text{J}$$

Demonstrates that work done is independent of the path taken in any electrostatic field.

Definition of Potential Difference and Potential

- The expression for the work *W* done by an external source in moving a charge *Q* from one point to another in an electric field E:

$$W = -Q\int_{\text{init}}^{\text{final}} \mathbf{E}\cdot d\mathbf{L}$$

- ***Potential difference V*** is defined as the work done by an external source in moving a unit positive charge from one point to another in an electric field:

$$\text{Potential difference} = V = -\int_{\text{init}}^{\text{final}} \mathbf{E}\cdot d\mathbf{L}$$

- Potential difference is measured in **joules per coulomb (J/C).** However, **volt (V)** is defined as a more common unit.

- The electric field intensity is then expressed in terms of the potential as:

$$\mathbf{E} = -\left.\frac{dV}{dL}\right|_{\text{max}} \mathbf{a}_N$$

Definition of Potential Difference and Potential

- For a point charge, we can find the potential difference between points *A* and *B* at radial distance r_A and r_B, choosing an origin at Q:

$$\mathbf{E} = E_r \mathbf{a}_r = \frac{Q}{4\pi\varepsilon_0 r^2}\mathbf{a}_r$$

$$d\mathbf{L} = dr\mathbf{a}_r$$

$$\begin{aligned} V_{AB} &= -\int_B^A \mathbf{E}\cdot d\mathbf{L} \\ &= -\int_{r_B}^{r_A} \frac{Q}{4\pi\varepsilon_0 r^2} dr \\ &= \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_A} - \frac{1}{r_B}\right) \end{aligned}$$

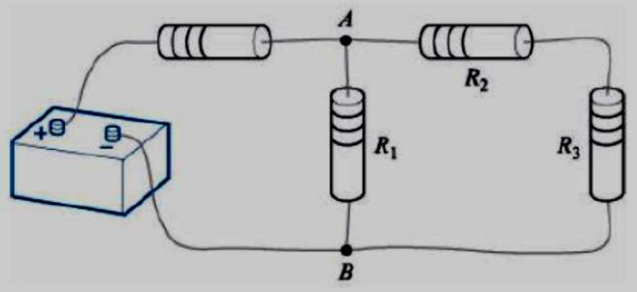
- Generally,

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

- Equipotential surface is a surface composed of all those points having the same value of potential.
- No work is involved in moving a charge around on an equipotential surface.

■ For *static fields*, no work is done in carrying the unit charge around any *closed path*.

$$\oint \mathbf{E} \cdot d\boldsymbol{L} = 0$$



- Potential conservation in a simple dc-circuit problem in the form of Kirchhoff's voltage law

$$\mathbf{E} = -\frac{dV}{dN}\mathbf{a}_N = -\text{grad } V$$

$$\mathbf{E} = -\nabla V$$

$$\mathbf{E} = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right)$$

- **Note: Gradient of a scalar is a vector.**

$$\nabla V = \frac{\partial V}{\partial \rho}\mathbf{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi + \frac{\partial V}{\partial z}\mathbf{a}_z$$

Cylindrical

$$\nabla V = \frac{\partial V}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi$$

Spherical

■ **Example**

Given the potential field, $V = 2x^2y - 5z$, and a point $P(-4,3,6)$, find V, E, direction of E, D, and ρ_V.

$$V_P = 2(-4)^2(3) - 5(6) = \underline{66 \text{ V}}$$

$$\mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right) = -4xy\mathbf{a}_x - 2x^2\mathbf{a}_y + 5\mathbf{a}_z$$

$$\mathbf{E}_P = -4(-4)(3)\mathbf{a}_x - 2(-4)^2\mathbf{a}_y + 5\mathbf{a}_z = \underline{48\mathbf{a}_x - 32\mathbf{a}_y + 5\mathbf{a}_z \text{ V/m}}$$

$$\mathbf{a}_{E,P} = \frac{\mathbf{E}_P}{|\mathbf{E}_P|}$$

$$\mathbf{D}_P = \varepsilon_0\mathbf{E}_P = \underline{425\mathbf{a}_x - 283.3\mathbf{a}_y + 44.27\mathbf{a}_z \text{ pC/m}^3}$$

$$\rho_v = \text{div } \mathbf{D} = \text{div } \varepsilon_0\mathbf{E} = (8.854\times10^{-12})(-4y) = -35.42y \text{ pC/m}^3$$

$$\text{At } P,\ \rho_v = -35.42(3) \text{ pC/m}^3 = \underline{-106.26 \text{ pC/m}^3}$$

D4.8. Given the potential field in cylindrical coordinates, $V = \frac{100}{z^2+1}\rho\cos\phi$ V, and point P at $\rho = 3$ m, $\phi = 60°$, $z = 2$ m, find values at P for (*a*) V; (*b*) **E**; (*c*) E; (*d*) dV/dN; (*e*) $\mathbf{a}_N$; (*f*) ρ_v in free space.

Ans. (*a*) 30.0 V; (*b*) $-10.00\mathbf{a}_\rho + 17.3\mathbf{a}_\phi + 24.0\mathbf{a}_z$ V/m; (*c*) 31.2 V/m; (*d*) 31.2 V/m; (*e*) $0.32\mathbf{a}_\rho - 0.55\mathbf{a}_\phi - 0.77\mathbf{a}_z$; (*f*) -234 pC/m^3

$$\nabla V = \frac{\partial V}{\partial \rho}\mathbf{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi + \frac{\partial V}{\partial z}\mathbf{a}_z$$

Cylindrical

$$\nabla V = \frac{\partial V}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi$$

Spherical


Thank
you